

Alex J. Hart

Aurora, Illinois

EDUCATION

Iowa State University; Ames, Iowa

Graduate Certificate, Power Systems Engineering, 2027

Northern Illinois University; DeKalb, Illinois

Master of Science, Electrical Engineering, 2023

Northern Illinois University; DeKalb, Illinois

Bachelor of Science, Electrical Engineering, 2022

WORK EXPERIENCE

Northrop Grumman; Rolling Meadows, Illinois

Systems Engineer

September 2023 – Present

- Built simulation models to guide engineering decisions
- Defined system specifications and interface requirements to establish baseline designs
- Collaborated with cross-disciplinary teams to execute system integration
- Coordinated technical design reviews across teams to ensure compliance with safety standards

Jet Propulsion Laboratory; Pasadena, California

Electrical Engineer Intern

June 2023 – August 2023

- Developed a telemetry system to monitor air quality in crew cabins

Electrical Engineer Intern

June 2022 – October 2022

- Researched tracking algorithms to detect residential gas leaks pt. II

Northern Illinois University; DeKalb, Illinois

Graduate Research Assistant (EE dept.)

November 2022 – May 2023

- Researched tracking algorithms to detect residential gas leaks

Student Helper (EE dept.)

June 2021 – August 2021

- Helped process simulation results with basic Python

PROJECTS

Transmission system study, 15-bus (personal)

June 2026 - Present

- Modeled a 15-bus urban/rural system in PSSE v36 to assess performance under load growth scenarios
- Conducted power flow, N-1 contingency, fault current, and transient stability analyses to identify grid violations and determine necessary transmission upgrades

Wind power plant study (academic)

May 2026

- Modeled a 2MW DFIG wind turbine in PLECS, with pitch control and power electronic converters to analyze generator response and real/reactive power flow to the grid
- Processed multi-year meteorological data from a wind energy laboratory in MATLAB to model wind generation profiles and forecast long-term resource capacity

Transmission system study, 30-bus (academic)

December 2025

- Modeled a 30-bus system in MATLAB to evaluate voltage profiles and line overloads under single-element outages
- Formulated OPF and Security-Constrained OPF models to determine dispatch strategies that avoid violations during contingencies

ACTIVITIES

NCEES FE ECE Exam

June 2026 – Present

- Preparing for the NCEES FE ECE exam scheduled for August 31, 2026

Abbott Power Plant / Solar Farm 2.0

July 2026

- Self-funded visit to the University of Illinois Abbott Power Plant and Solar Farm 2.0 to learn about how these unique generation facilities provide power to 60,000 students and faculty on campus

Power and Energy Conference at Illinois (PECI)

April 2026

- Self-funded visit to PEGI 2026 to learn about emerging research in power and energy